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Started Thu, 10 Sep

Solving nonlinear equations

**1.2: Interval Bisection**

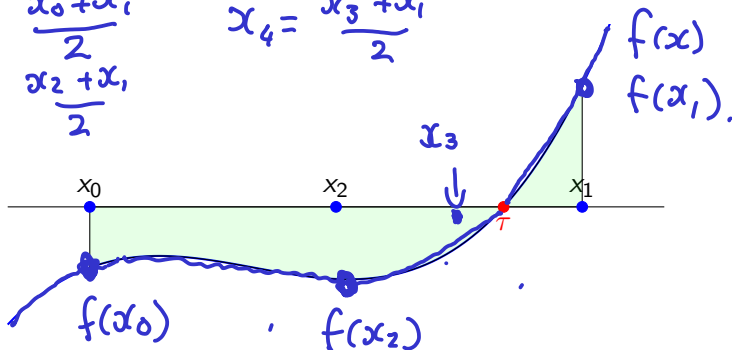
MA385 – Numerical Analysis

September 2026

$$x_2 = \frac{x_0 + x_1}{2}$$

$$x_3 = \frac{x_2 + x_1}{2}$$

$$x_4 = \frac{x_3 + x_1}{2}$$



# 0. Outline

1 Bisection

2 The bisection method works

3 Improving upon bisection

4 Exercises

For more details, see Section 1.6 (The Bisection Method) of Süli and Mayer, *An Introduction to Numerical Analysis*

# 1. Bisection

The most elementary algorithm is the “*Bisection Method*” (also known as “Interval Bisection”). Suppose that we know that  $f$  changes sign on the interval  $[a, b] = [x_0, x_1]$  and, thus,  $f(x) = 0$  has a solution,  $\tau$ , in  $[a, b]$ . Proceed as follows

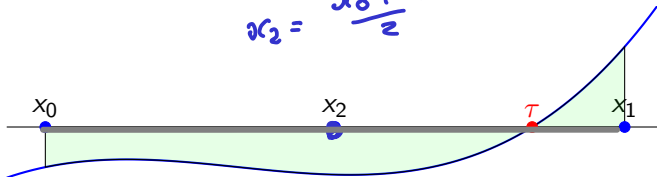
1. Set  $x_2$  to be the midpoint of the interval  $[x_0, x_1]$ .
2. Choose one of the sub-intervals  $[x_0, x_2]$  and  $[x_2, x_1]$  where  $f$  change sign;
3. Repeat Steps 1–2 on that sub-interval, until  $f$  is sufficiently small at the end points of the interval.

Notes: by convention  $[x_i, x_j]$  means  
 $x_i \leq x \leq x_j$  if  $x_i < x_j$ . otherwise  
it is  $x_j \leq x \leq x_i$

# 1. Bisection

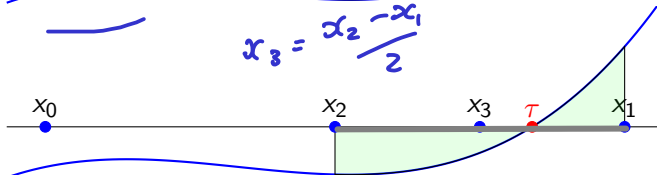
Step 1

$$x_2 = \frac{x_0 + x_1}{2}$$



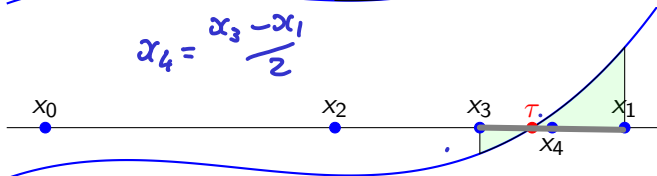
Step 2

$$x_3 = \frac{x_2 - x_1}{2}$$



Step 3

$$x_4 = \frac{x_3 - x_1}{2}$$



# 1. Bisection

This may be expressed more precisely using some *pseudocode*.

## The Bisection Algorithm

Set  $\epsilon$  to be the stopping criterion.

If  $|f(a)| \leq \epsilon$ , return  $a$ . Exit.

If  $|f(b)| \leq \epsilon$ , return  $b$ . Exit.

Set  $x_L = a$  and  $x_R = b$ .

Set  $k = 1$

while(  $|f(x_k)| > \epsilon$  )

$x_{k+1} = (x_L + x_R)/2$ ;

    if (  $f(x_L)f(x_{k+1}) < 0$  )

$x_R = x_{k+1}$ ;

    else

$x_L = x_{k+1}$

    end if;

$k = k + 1$

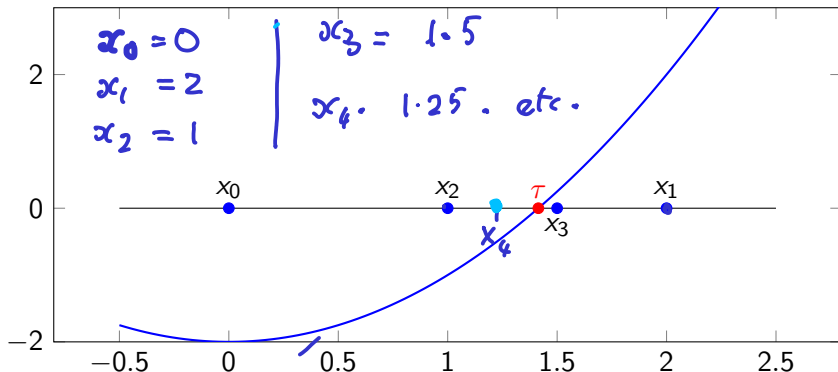
end while;

# 1. Bisection

## Example

Find an estimate for  $\sqrt{2}$  that is correct to 6 decimal places.

**Solution:** Use bisection to solve  $f(x) := x^2 - 2 = 0$  on the interval  $[0, 2]$ .



# 1. Bisection

## Example

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**Solution:** Use bisection to solve  $f(x) := x^2 - 2 = 0$  on the interval  $[0, 2]$ .

| $k$      | $x_k$    | $ f(x_k) $ | $ \tau - x_k $ |
|----------|----------|------------|----------------|
| 0        | 0.000000 | 2.00e+00   | 1.41e+00       |
| 1        | 2.000000 | 2.00e+00   | 5.86e-01       |
| 2        | 1.000000 | 1.00e+00   | 4.14e-01       |
| 3        | 1.500000 | 2.50e-01   | 8.58e-02       |
| 4        | 1.250000 | 4.38e-01   | 1.64e-01       |
| 5        | 1.375000 | 1.09e-01   | 3.92e-02       |
| 6        | 1.437500 | 6.64e-02   | 2.33e-02       |
| 7        | 1.406250 | 2.25e-02   | 7.96e-03       |
| 8        | 1.421875 | 2.17e-02   | 7.66e-03       |
| 9        | 1.414062 | 4.27e-04   | 1.51e-04       |
| $\vdots$ | $\vdots$ | $\vdots$   | $\vdots$       |
| 22       | 1.414214 | 1.62e-06   | 5.72e-07       |
| 23       | 1.414214 | 2.69e-07   | 9.50e-08       |

$$(1/2)^k |x_1 - x_0|$$

## 2. The bisection method works

One of the main advantages of the Bisection method is that it will always work, providing only that  $f$  is continuous on  $[a, b]$ , and that the solution exists. Furthermore, we can prove...

### Theorem 1.2.1

Let  $x_k$  be the  $k$ th iteration generated by the Bisection Method. Then

$$|\tau - x_k| \leq \left(\frac{1}{2}\right)^{k-1} |b - a|, \quad \text{for } k = 2, 3, 4, \dots$$

*Proof:* Clearly, since  $a = x_0$ ,  $b = x_1$ ,  
and  $a \leq \tau \leq b$ , we have that  
 $|\tau - x_0| \leq |b - a|.$  ✓

## 2. The bisection method works

Suppose at step  $k$  that

$$|\tau - x_k| \leq \left(\frac{1}{2}\right)^{k-1} |b-a|.$$

At step  $k+1$ , we bisect the interval again, giving

$$\begin{aligned} |\tau - x_{k+1}| &\leq \frac{1}{2} \left[ \left(\frac{1}{2}\right)^{k-1} |b-a| \right] \\ &= \left(\frac{1}{2}\right)^k |b-a|. \end{aligned}$$

So, by induction, the proof follows.

### 3. Improving upon bisection

A disadvantage of bisection is that it is not particularly efficient. So our next goal will be to derive better methods, particularly the **Secant Method** and **Newton's method**. We also have to come up with some way of expressing what we mean by **"better"**.

Finished here 4pm, Thursday 10 Sep (Week 1).

## 4. Exercises

### Exercise 1.2.1

Suppose we want to find  $\tau \in [a, b]$  such that  $f(\tau) = 0$  for some given  $f$ ,  $a$  and  $b$ . Write down an estimate for the number of iterations  $K$  required by the bisection method to ensure that, for a given  $\varepsilon$ , we know  $|x_k - \tau| \leq \varepsilon$  for all  $k \geq K$ . In particular, how does this estimate depend on  $f$ ,  $a$  and  $b$ ?

### Exercise 1.2.2

How many (decimal) digits of accuracy are gained at each step of the bisection method? (If you prefer, how many steps are needed to gain a single (decimal) digit of accuracy?)

## 4. Exercises

### Exercise 1.2.3

Let  $f(x) = e^x - 2x - 2$ . Show that there is a solution to the problem: *find  $\tau \in [0, 2]$  such that  $f(\tau) = 0$ .*

Taking  $x_0 = 0$  and  $x_1 = 2$ , use 6 steps of the bisection method to estimate  $\tau$ . You may use a computer program to do this, but please note that in your solution.

Give an upper bound for the error  $|\tau - x_6|$ .

## 4. Exercises

### Exercise 1.2.4

We wish to estimate  $\tau = \sqrt[3]{4}$  numerically by solving  $f(x) = 0$  in  $[a, b]$  for some suitably chosen  $f$ ,  $a$  and  $b$ .

- (i) Suggest suitable choices of  $f$ ,  $a$ , and  $b$  for this problem.
- (ii) Show that  $f$  has a zero in  $[a, b]$ .
- (iii) Use 6 steps of the bisection method to estimate  $\sqrt[3]{4}$ .  
You may use a computer program to do this, but please note that in your solution.
- (iv) Use Theorem 7 to give an upper bound for the error  $|\tau - x_6|$ .